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Economic Evaluation

Quantifying the Value of Reduced Health Disparities: Low-Dose Computed Tomography Lung Cancer Screening of High-Risk Individuals Within the United States

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ABSTRACT

Objective: This study aimed to measure the value of increasing lung cancer screening rates for high-risk individuals and its impact on health disparities.

Methods: The model estimated changes in health economic outcomes if low-dose computed tomography screening increased from current to 100% compliance, following clinical guidelines. Current low-dose computed tomography screening rates were estimated by income, education, and race, using 2017-2019 Behavioral Risk Factor Surveillance System data. The model contained a decision tree module to segment the population by screening outcomes and a Markov chain module to estimate cancer progression over time. Model parameters included information on survival, quality of life, and costs related to cancer diagnosis, treatment, and adverse events. Distributional cost-effectiveness analysis estimated the net monetary value from reduced health disparities—measured using quality-adjusted life expectancy—across income, education, and race groups. Outcomes were assessed over 30 years.

Results: Lung cancer screening eligibility using US Preventive Services Task Force guidelines was higher for individuals with income <$15 000 (47.2%) and without a high-school education (46.1%) than individuals with income >$50 000 (16.6%) and with a college degree (13.5%), respectively. Increasing lung cancer screening to 100% compliance was cost-effective ($64 654 per quality-adjusted life-year) and produced economic value by up to $560 per person ($182.1 billion for United States overall). Up to 32.2% of the value was due to reductions in health disparities.

Conclusions: Significant value in increasing lung cancer screening rates derived from reducing health disparities. Policy makers and clinicians may not be appropriately prioritizing cancer screening if value from reducing health disparities is unconsidered.

Keywords: cancer screening, distributional cost-effectiveness modeling, health disparities, lung cancer.

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Introduction

Lung cancer is the leading cause of cancer death in the United States with an annual mortality rate of 36.7 per 100 000 people (2015-2019).[1] Lung cancer imposes significant economic burden: out-of-pocket costs for lung cancer treatment were $1.35 billion in 2019 and lost earnings due to lung cancer were $21.3 billion in 2015.[2,3] Despite the advent of numerous therapies to treat lung cancer in recent years leading to improvements in survival,[4] identifying lung cancer at an early stage still remains pivotal for maximizing patient health outcomes.[5]

Clinical guidelines recommend annual lung cancer screening using low-dose computed tomography (LDCT) among high-risk individuals. LDCT screening decreases lung cancer mortality by detecting cancer at an earlier, more curable, stage.[6,7] The 2013 US Preventive Services Task Force (USPSTF) recommended annual LDCT screening for individuals aged ≥55 years with a ≥30 pack-year smoking history and currently smoking or quit smoking within the past 15 years; in 2021 USPSTF updated their recommendation to (1) lower the screening age from 55 to 50 years and (2) lower smoking history threshold from 30 to 20 pack-years.[8,9]

Although effective in improving clinical outcomes and decreasing lung cancer mortality,[6,10,11] LDCT screening compliance is low: between 3% and 14% of eligible adults undergo LDCT screening.[12-14] In addition, demographic or socioeconomic status (SES) affects screening uptake. For example, Black Americans are less likely to receive cancer screening.[7,15] Moreover, smoking rates remain higher for individuals of lower income and education,[16] and individuals from the lowest SES group had a 13% higher rate of lung cancer mortality than their higher SES counterparts.[17]

Health equity has emerged as a key "pillar" of the Centers for Medicare and Medicaid Services' (CMS) long-term strategic vision,

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including actions such as closing gaps in healthcare quality, access, outcomes, and expanding the use of health equity data.[18] Distributional cost-effectiveness analysis (DCEA) modeling allows researchers to estimate the impact of intervention while taking into account of health disparities between affluent and disadvantaged groups. DCEA leverages societal preferences for reduced health disparities and assigns more value—relative to traditional cost-effectiveness approaches—to interventions where the disease of interest disproportionately affects disadvantaged groups or the intervention of interest is differentially beneficial for disadvantaged groups.[19] Despite a growing DCEA literature on health technologies and rare conditions, such as sickle cell disease,[19,20] the impact of health disparities from increased screening intervention for individuals of high risk of lung cancer is not well understood.

Against this backdrop, this study uses a DCEA approach to quantify the value of increasing LDCT screening rates and measure its impact on health disparities across income, education, and race/ethnicity groups.

Method

Model Approach

We developed a decision analytic model to quantify the value of increasing current LDCT screening rates to 100%, taking into account the societal value of reduced health disparities over a 30-year time frame (ie, lifetime). The study methodology relied on 4 key steps: (1) estimate current LDCT screening eligibility and uptake, (2) measure health disparities across different income, education, and race/ethnicity groups, (3) quantify the economic value of LDCT screening, and (4) measure the value of increased LDCT screening from reduced health disparities. This study followed the Consolidated Health Economic Evaluation Reporting Standards 2022 checklist, which can be located in the Appendix in Supplemental Materials found at https://doi.org/10.1016/j.jval.2023.12.014.[21] Outcomes were assessed over 30 years.

Eligibility and uptake of LDCT screening

Using United States–representative data from the 2017-2019 Behavioral Risk Factor Surveillance System (BRFSS), we estimated variation in current eligibility and uptake of LDCT screening rates among eligible individuals across income, education, and race/ethnicity subgroups.[14,22,23] Race/ethnicity categories were non-Hispanic White, non-Hispanic Black, Hispanic, and other races (defined as Asian, Indian American, Native Hawaiian, multiracial, and other races). Individuals included in the sample were (1) aged 50 to 79 years, (2) had a ≥20 pack-year smoking history, and (3) currently smoked or had quit within the past 15 years, closely matching the eligibility criteria for the 2021 USPSTF lung cancer screening guidelines.[8]

Disparities in health outcomes

Variations in health outcomes at baseline across income, education, and race/ethnicity were measured using the quality-adjusted life expectancy (QALE) metric. QALE combines life expectancies and quality of life of a general population of interest (see Appendix Table 1 in Appendix B in Supplemental Materials found at https://doi.org/10.1016/j.jval.2023.12.014 for calculations).[19,24] Traditional cost-effective analysis quantifies increases in health outcomes using incremental quality-adjusted life-years (QALYs), which assesses the additional life-years patients gain from an intervention and adjusting for quality of life.[25] However, in DCEA, QALE is used because disparities in health outcomes are measured based on the social welfare rather than utility of average patients.[19]

Estimation of economic value of full LDCT screening

To estimate the value of LDCT screening, a cost-effective analysis model was developed using Microsoft Excel. The model follows closely previously published models that have estimated the economic value of LDCT screening using traditional cost-effectiveness methods.[25,26] The model was composed of a decision tree for screening procedures with branches that describe cancer progression using a first-order Markov chain with a 1-month cycle length and 3 severity stages of lung cancer—stages I/II, III, and IV—and death (Fig. 1). The bottom left branch contains a Markov chain that estimates the long-term outcomes for patients who are diagnosed of lung cancer and treated with the standard of care—hereafter, the long-term outcomes module. The bottom right branch contains a Markov chain that estimates the natural progression of lung cancer in the absence of cancer diagnosis and treatment—hereafter, the delay-period module. Cancer progresses faster for those who are undiagnosed and untreated than those diagnosed and treated. Undiagnosed and untreated patients will continue to undergo lung cancer screening annually, until they receive a true positive test result, matching the USPSTF guidelines.[8] The primary benefit of increased screening is an earlier diagnosis and initiation of cancer treatment.

In the full compliance arm, LDCT screening accuracy is based on the American College of Radiology Lung CT Screening Reporting & Data System guidelines.[27] Among those in the comparison arm who are not screened but who have cancer, the disease progresses based on the "natural progression" model in that month. These patients with undiagnosed cancer are eligible to receive LDCT screening in subsequent months and be diagnosed and move to the long-term outcomes model. The economic value of increasing the screening rate to 100% was measured based on incremental cost-effectiveness ratio and net health benefit.

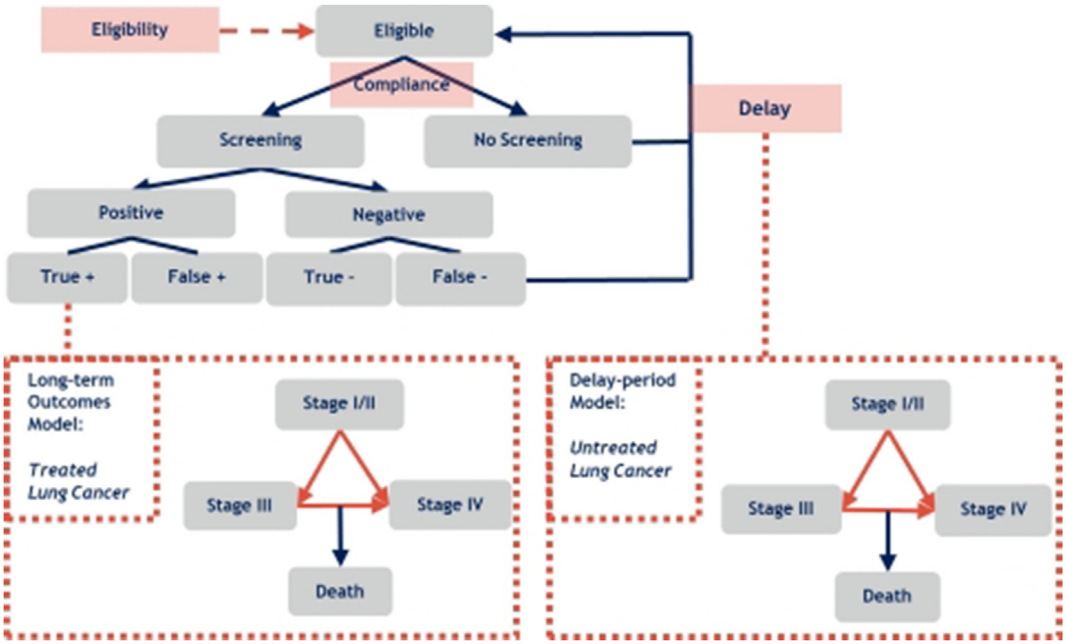
Estimation of societal value of full LDCT screening considering reduced health disparities

The societal value of the intervention was assessed by equally distributed equivalent (EDE) QALY, which is the level of population health in an equal distribution that yields an equivalent amount of social welfare to the distribution after increased LDCT screening was implemented.[19] Conceptually, EDE QALY estimates not only the net economic value of the intervention of interest but the value that the treatment brings to society through health disparities reductions.

We calculated EDE QALY in 4 steps. First, changes in QALYs and costs made by the intervention were derived for each subgroup (ie, income, education, race/ethnicity) as described earlier. Second, net health benefit, defined as gains in health net of opportunity costs incurred by resources used, was derived for each subgroup. Third, for each subgroup, we estimated QALE by adding the subgroup-specific net health benefit to the baseline subgroup-specific QALE from current screening rates. Finally, we converted the set of subgroup-specific QALEs into EDE QALYs attained by full LDCT screening, using Atkinson (relative) or Kolm (absolute) inequality aversion index, set to 10.95 and 0.15, respectively[19] (see and Appendix Table 1 in Appendix B in Supplemental Materials found at https://doi.org/10.1016/j.jval.2023.12.014 for detailed calculations).

Once the EDE QALYs were derived, we quantified the value gained from reduced health disparities. We first calculated incremental EDE QALYs (eider) by subtracting the baseline QALE (calculated based on the current screening rates) from EDE QALY

Figure 1. Model structure.



(calculated based on the full LDCT compliance). The baseline QALE was derived from the baseline subgroup-specific QALEs using the inequality aversion indices just as in EDE QALY. Second, we calculated the value created by reduced health disparities, incremental EDE net of total net health benefit. Values were monetized using the US societal willingness-to-pay threshold of $100 000.[28] We obtained the values toward the US economy by multiplying the estimate of the US population between 50 and 79 years old (104.7 million) to the individual monetary gains.[29] The share of value from reduced health disparities was also compared with previously published DCEAs.

Model Parameters

One principal driving factor explaining the value of full LDCT compliance is the difference of lung cancer progression rates between the long-term outcomes and the delay-period modules. The progression rates in the long-term outcomes module were derived based on the survival rates of patients with lung cancer from 2019 Surveillance, Epidemiology, and End Results Program under the exponential survival assumption.[30] The progression rates in the delay-period module were calculated using the quarterly rate of natural progression of patients with undiagnosed lung cancer from literature, which estimates the disease progression of a high-risk population—similar to those eligible for LDCT screening—based on epidemiological data.[31] The sensitivity and specificity of LDCT were based on estimates from the American College of Radiology Lung CT Screening Reporting & Data System, which were calculated to be 84.93% and 87.19%, respectively.[27] We used the 2019 Surveillance, Epidemiology, and End Results Program registry to identify the distribution of the lung cancer stage in the initial month.[32] Lung cancer diagnosis, treatment, and adverse events costs included in the model were adjusted for inflation to 2022 US dollar.[33-37] Indirect costs included nonmedical out-of-pocket cost, caregiver costs, nonhealthcare sector costs, and patient productivity costs, which were also adjust for inflation to 2022 US dollar.[38-42] All outcomes used a discount rate of 3%, matching standard cost-effectiveness guidelines.[43] See Table 1 for a condensed list of model parameters and Appendix Table 2 in Appendix C in Supplemental Materials found at https://doi.org/10.1016/j.jval.2023.12.014 for all model parameters.

Results

LDCT Screening Eligibility and Screening Rate

Among those aged 50 to 79 years, 25.2% were eligible to receive LDCT screening for lung cancer based on the 2021 USPSTF guidelines (Fig. 2).[8] Individuals from lower income or education subgroups had higher eligibility rates than those from higher income or education subgroups. Eligibility decreased monotonically as income level or education attainment increased, from 47.2% for individuals with income of <$15 000 to 16.6% for those with income of >$50 000 and from 46.1% for individuals without a high-school degree to 13.5% college degrees holders. Non-Hispanic White, non-Hispanic Black, Hispanic, and other race individuals had eligibility rates of 24.7%, 26.6%, 20.4%, and 32.9%, respectively.

Among eligible individuals, only 12.5% self-reported receiving LDCT screening for lung cancer (Fig. 2), similar to previous estimates in the literature using BRFSS data.[14] Screening rates among those meeting USPSTF eligibility criteria varied across income (ranging between 10.9% and 14.7%), education (11.9% and 14.5%), and race/ethnicity (8.5% and 13.7%) subgroups.

Health Disparities

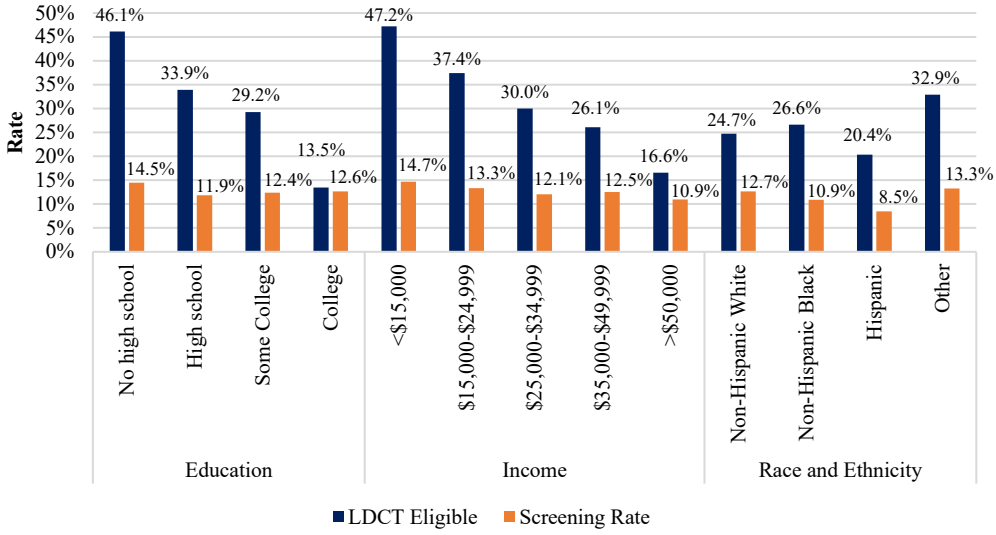
Individuals with lower income or education attainment, as well as racial/ethnic minority individuals, had worse health outcomes, in terms of QALE at baseline (Table 2). Baseline subgroup-specific QALEs at age 50 years were 18.774 and 23.807 for the <$15 000 and >$50 000 income groups and 18.776 and 22.471 for the no high-school and college education groups, respectively. Among the groups based on race/ethnicity, baseline QALE was lower for non-

Table 1. Condensed table of inputs.

Parameter category	Parameter	Parameter value	Source
Clinical inputs/distributions	Distribution of lung cancer stages in the initial period		
	Stage I/II	29.66%	Author calculations based on NIH SEER (2019)[30]
	Stage III	20.00%	Author calculations based on NIH SEER (2019)[30]
	Stage IV	50.34%	Author calculations based on NIH SEER (2019)[30]
	Transition probabilities (1-month cycle)		
	Long-term outcomes model		
	From stage I/II to stage III	0.94%	Author calculations based on NIH SEER (2019)[30]
	From stage I/II to stage IV	0.35%	Author calculations based on NIH SEER (2019)[30]
	From stage I/II to death	0.04%	Author calculations based on NIH SEER (2019)[30]
	From stage III to stage IV	2.88%	Author calculations based on NIH SEER (2019)[30]
	From stage III to death	0.31%	Author calculations based on NIH SEER (2019)[30]
	From stage IV to death	9.06%	Author calculations based on NIH SEER (2019)[30]
	Delay-period model		
	From stage I/II to stage III	5.08%	Author calculations based on Hofer et al[31]
	From stage I/II to stage IV	3.74%	Author calculations based on Hofer et al[31]
	From stage I/II to death	4.86%	Author calculations based on Hofer et al[31]
	From stage III to stage IV	5.11%	Author calculations based on Hofer et al[31]
	From stage III to death	5.37%	Author calculations based on Hofer et al[31]
	From stage IV to death	11.12%	Author calculations based on Hofer et al[31]
	LDCT test outcomes		
	Sensitivity	84.93%	Pinsky et al[27]
	Specificity	87.19%	Pinsky et al[27]
Quality of life	Baseline quality of life by lung cancer stage		
	Stage I/II	0.825	Sturza et al[44]
	Stage III	0.772	Sturza et al[44]
	Stage IV	0.573	Sturza et al[44]
Cost inputs	LDCT screening	$147	CMS,[37] ACR[45]
	Monthly treatment cost of lung cancer		
	Long-term outcomes model		
	Stage I/II	$875	Author calculations based on Sheehan et al[35]
	Stage III	$3110	Author calculations based on Sheehan et al[35]
	Stage IV	$5120	Author calculations based on Sheehan et al[35]

ACR indicates American College of Radiology; CMS, Centers for Medicare and Medicaid Services; LDCT, low-dose computed tomography; NIH, National Institutes of Health; SEER, Surveillance, Epidemiology, and End Results.

Figure 2. LDCT screening eligibility under 2021 USPSTF guidelines and screening rates by subgroup, 2017-2019.



LDCT indicates low-dose computed tomography; USPSTF, US Preventive Services Task Force.

Hispanic Black individuals (21.336) relative to non-Hispanic White (22.489), Hispanic (22.287), and other races (23.092).

Economic Value of Full LDCT Screening

The model predicted that increasing LDCT screening rates from current compliance to 100% resulted in 0.69 more screenings per eligible individual-year (0.10 vs 0.79) (Appendix Fig. 1 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2023.12.014). Life-years increased by 0.027 (0.18-0.45) as a result of more screenings, earlier diagnosis, and receiving treatment at an earlier stage. At a 3% discount rate, the increased LDCT compliance produced 0.1327 higher QALYs (14.1191 vs 13.9864) and $8579 higher costs ($12 242 vs $3663) for an eligible individual (Appendix Figs. 2 and 3 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2023.12.014). Most health gains (measured in QALYs) and changes in cost were generated during the treatment period among patients diagnosed of lung cancer. The estimated incremental cost-effectiveness ratio—$64 654 per QALY—fell below the $100 000 per QALY threshold (Appendix Table 3 in Appendix D in Supplemental Materials found at https://doi.org/10.1016/j.jval.2023.12.014).[28]

Societal Value of Full LDCT Screening Considering Reduced Health Disparities

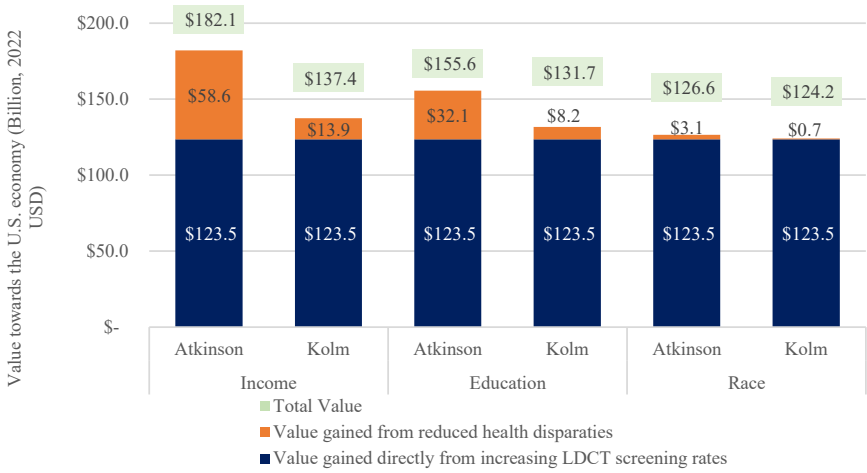
Increasing LDCT screening rates to full compliance improved health outcomes for all individuals, but particularly disadvantaged

Table 2. QALE by subgroup.

Group		QALE		
		Baseline (current screening rate)	Intervention (full LDCT compliance)	Difference (intervention-baseline)
Income	<$15 000	18.774	18.796	0.022
	$15 000-$24 999	20.499	20.516	0.017
	$25 000-$34 999	21.583	21.598	0.015
	$35 000-$49 999	22.471	22.484	0.013
	>$50 000	23.807	23.815	0.008
Education	No high school	18.776	18.797	0.021
	High school	20.498	20.514	0.016
	Some college	21.581	21.595	0.014
	College	22.471	22.478	0.007
Race	Non-Hispanic White	22.489	22.500	0.011
	Non-Hispanic Black	21.336	21.354	0.018
	Hispanic	22.287	22.288	0.001
	Other	23.092	23.104	0.012
Weighted average		22.440	22.451	0.011

LDCT indicates low-dose computed tomography; QALE, Quality-adjusted life expectancy.

Figure 3. Values toward the US economy by increasing LDCT screening from current to 100% compliance under 2021 USPSTF guidelines.



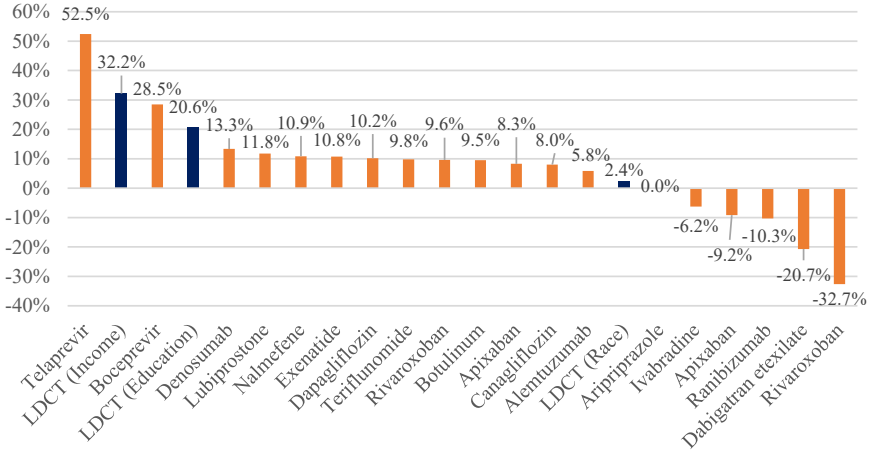
LDCT indicates low-dose computed tomography; USPSTF, US Preventive Services Task Force.

populations (ie, lower income, lower education, and racial/ethnic minority individuals) (Table 2). QALE increased by 0.022 and 0.008 for individuals who earn <$15 000 and >$50 000 annually (18.796 and 23.815 after intervention) and by 0.021 and 0.007 for individuals with no high-school and college education (18.797 and 22.478 after intervention), respectively. For race/ethnicity, QALE increased more for non-Hispanic Black individuals (0.018) than for individuals that were non-Hispanic White (0.011), Hispanic (0.001), or an other race (0.012) (21.354, 22.500, 22.288, and 23.104 after intervention for non-Hispanic Black, non-Hispanic White, Hispanic, and other races, respectively).

A large share of the societal value gained from increased LDCT screening was explained by the reduction in health disparities (Fig. 3). EDE QALYs of the intervention derived using Atkinson social welfare function were 0.0174, 0.0149, and 0.0121 for income, education, and racial groups, respectively, corresponding to total monetary values toward society over a 30-year period of $182.1 billion ($1740 per person), $155.6 billion ($1486 per person), and $126.6 billion ($1209 per person). Of the total values, 32.2% ($58.6 billion; $560 per person), 20.6% ($32.1 billion; $307 per person), and 2.4% ($3.1 billion; $29 per person) were explained by reduced health inequalities for income, education, and racial groups, respectively. The remaining values were due to the health benefit of screening to the average patient using standard cost-effectiveness analysis. Qualitatively similar results were obtained when the Kolm's social welfare utility function was applied, with smaller magnitude in the share of value due to reduced health disparities.

This study also considered alternative scenarios with more modest increases in target screening levels and varying levels of inequality aversion. In particular, increasing LDCT screening to 20%, 40%, 60%, and 80% of USPSTF recommendations resulted in similar proportions of the total value came from reduced health disparities were uncovered (Appendix Table 4 in Appendix E in Supplemental Materials found at https://doi.org/10.1016/j.jval.2023.12.014). Additional sensitivity analysis included changing the inequality aversion parameter (ie, societal tolerability of inequality) by ±10%, and using alternative inequality aversion parameters from the literature[46] had a more substantial impact on the results, but incorporating the impact on health disparities always increased the value of LDCT screening relative to approaches that did not take into account health disparities

Figure 4. Percent value from reduced disparities between Love-Koh et al's (2019) and this study (Atkinson).



(Appendix Figs. 4 and 5 in Appendix F in Supplemental Materials found at https://doi.org/10.1016/j.jval.2023.12.014).

Discussion

Compared with previous research, the share of value gained from reduced income and education disparities was higher for LDCT full compliance than most other intervention measured using the DCEA approach in the literature (Fig. 4; Appendix Tables 4 and 5 in Appendix F in Supplemental Materials found at https://doi.org/10.1016/j.jval.2023.12.014).[19] Furthermore, the value gained from reduced income and education disparities as a result of full LDCT compliance was also significantly higher than interventions for smoking cessation.[20,47] One reason the value of reduced health disparities is relatively high for LDCT screening is that LDCT eligibility is much higher among socially disadvantaged groups than other health interventions.

Despite the DCEA literature expanding for other conditions, such as sickle cell disease,[46] this study is the first to quantify the economic value of LDCT screening for lung cancer in terms of its impact on health disparities. Disadvantaged individuals—particularly individuals who had lower income, were less educated, or were Black Americans—were more likely to be eligible for LDCT screening: eligibility was higher than the overall eligibility (25.2%) by 22.0%, 20.9%, and 1.4% for individuals with income of <$15 000, those without a high-school education, and non-Hispanic Black Americans, respectively. Therefore, increased rates of LDCT screening from current levels (12.5% of those eligible) would disproportionately benefit disadvantaged populations and result in significant value toward the US economy created by reduced health disparities. In particular, increasing LDCT screening from current levels to 100% compliance produced up to $182.1 billion toward the US economy and up to 32.2% of the total value ($58.6 billion) was attributed to reduced health disparities.

Although this study demonstrated the value of increased LDCT screening both to the average individual and in terms of societal benefit by reducing health disparities, increasing screening rates in practice may face a number of challenges. Clinicians often do not have access to data to identify individuals eligible for LDCT screening. Although clinicians collect information on tobacco history, data on smoking pack-years are less frequently recorded in electronic health records. In the future, approaches that allow for more detailed information on smoking history and allow for screening eligibility assessment should be evaluated across healthcare settings. Furthermore, payers and health systems could develop and adopt a lung cancer screening quality measure that assesses the LDCT screening uptake. The adoption of a quality measure may motivate payers and health systems to (1) identify disparities within their care network, (2) establish community health goals, and (3) incentivize improvements in health outcomes and become more efficient in care delivery.

Moreover, improved LDCT screening is important for numerous CMS initiatives. Reducing health disparities and closing gaps in healthcare quality, access, and outcomes through health equity initiatives has taken center stage as a key pillar of CMS' long-term strategic vision. One priority for CMS is to standardize and expand the collection and use of health equity data, including data elements related to race/ethnicity, SES, sexual orientation, gender identity, disability, and geography, to name a few. In particular, data on race/ethnicity and SES are needed to better track how LDCT screening outcomes change over time, to better target interventions to improve LDCT screening rates. This view is supported by the Office of the Inspector General, which notes that limited data on race/ethnicity, as well as other factors (eg, education, income), make it challenging to track changes in health disparities. Finally, as the Inflation Reduction Act mandates that the CMS begin negotiating a maximum fair price for selected drugs, the DCEA methodology offers a quantitative approach for incorporating the value of reduced health disparities in any value-based pricing negotiation.

This study has several limitations. First, the results of the study depend on the point estimate of inequality aversion from a single study. However, additional research on societal preferences for reducing health disparities is needed. Second, although BRFSS data provide a wide range of health outcomes data, individual states can elect to release select BRFSS optional module questions, such as lung cancer screening questions.[23] Variation in the states that elected to release these questions limited opportunity to generalize lung cancer screening figures nationally. Third, estimated baseline screening rate (12.5%) was higher than some previous studies (4%-6.5%) and core results assumed perfect LDCT screening compliance.[12,13] The reason for this discrepancy could be that BRFSS contains self-reported data on CT for lung cancer screening. In addition, Screening rates were based on USPSTF guidelines with the intention of increasing the proportion of minority patients eligible for screening. Due to this issue, the screening rates reported in this study may be underestimated. Fourth, this study did not consider disparities in uptake of LDCT screening among specific populations despite some recent evidence that LDCT screening varies by race among veterans.[48] Fifth, our study did not use a single measure of health disparities such as the Social Vulnerability Index. Although a single measure of social determinants of health is useful, it may make it more difficult for key decision makers to interpret. In addition, one cannot calculate Social Vulnerability Index using BRFSS due to missing variables, but BRFSS data were preferred by the Centers for Disease Control and Prevention as a source for measuring LDCT screening rates. Finally, the study followed previous LDCT cost-effectiveness analysis model structures and parameterizations; thus, the core contribution of this study should be seen as how the incorporation of valuing changes in health disparities affects estimated screening value.

Conclusion

LDCT lung cancer screening of high-risk individuals remains low (12.5% overall). However, this study found that increasing screening rates to 100% was cost-effective and reduced health disparities between affluent and disadvantaged groups. In particular, up to $182.1 billion could be added to the US economy if screening rates were to increase from current levels to 100%. Up to 32.2% of the value (or $58.6 billion) from increased LDCT screening rates was due to reduced health disparities. Using the DCEA methodology to incorporate health equity considerations into economic evaluations, policy makers, payers, and clinicians will be able to more appropriately prioritize screening technologies that have potential to significantly reduce health disparities.

Author Disclosures

Links to the disclosure forms provided by the authors are available here.

Supplemental Material

Supplementary data associated with this article can be found in the online version at https://doi.org/10.1016/j.jval.2023.12.014.

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